Drone Swarm Strategies for Wildfire Containment: An Agent-Based Modeling Study

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Abstract

Wildfires are increasing in frequency and severity due to climate change, leading to a growing strain on both fire detection and human firefighting capacity. Although autonomous drones seem promising as a potential tool to augment disaster response throughout the process, the effectiveness of decentralized drone strategies are difficult to empirically test. Moreover, drone responses are difficult to measure due to the significantly diverse microenvironments that forest fires have. In this study, I extend an existing agent-based forest fire model by introducing autonomous drone agents that detect and extinguish fires. Using local sensing rules and drone behavioral parameters, I investigate what patterns emerge under different environmental and drone settings, when and why baseline drone strategies fail, and whether alternative rules could mitigate these failures. Results illustrate how intermediate parameter configurations outperform the rest, and large detection radii can lead to a loss of coverage due to the tendency of ineffective convergence. Moreover, local sensing strategies are vulnerable to the exploitation-exploration trade-offs, leading to "flare-ups". These findings suggest the need for more efficient local sensing rules before implementation.

Introduction

Forest fires are uncontrolled burning processes that could lead to serious ecological and economic damage. There are a large number of information systems to model the behavior of wildfires, such as the BehavePlus system and WFDS, but little research has covered the firefighting effectiveness. As fires grow in intensity and size, human-led efforts start to face growing resource constraints, caused by limitations in situational awareness and coordination. One emerging technology in this area is autonomous drone swarms, with more startups and nonprofits working on the detection, monitoring, and suppression of wildfires.

However, despite a growing interest in deploying this technology, large-scale experiments are costly and dangerous. As a result, failure modes and edge cases are significantly underexplored. Hence, Agent-based modeling (ABM) is a powerful alternative that allows researchers and policy makers to test decentralized strategies and average out the outcome under controlled and realistic conditions to reveal emergent outcomes that are logistically impossible to test.

In this paper, I use agent-based modeling to explore the effectiveness and limitations of autonomous drone swarms in forst fire detection and suppression. More specifically, I extend an existing forest fire model to include drone agents that detect and extinguish fires based on local sensing rules. To understand how decentralized drone behavior interact with forest fire dynamics, this paper will explore the following research questions:

- What patterns emerge under different flammability, wind, and drone settings?
- When and why does the baseline drone strategy fail?
- Can alternative rules address these failures?

Methods

Base Model: Forest Fire Model

I used a NetLogo forest fire model that simulates the spread of fire through a forested landscape as the base model. The environment is a two-dimensional grid populated by trees and fires. Fires spread by scanning their surroundings within a specified radius, igniting then killing nearby trees while generating new fire agents. A fire extinguishes if it finds no trees within its radius.

Depending on the modes and initial parameter settings, the model exhibits varying fire behaviors, including spread patterns, runaway fires, and even extinctions (though it is highly unlikely). The probability that the fire is contained is also affected by parameters such as wind, terrain, and a "big jump" mechanism that represents ember transport. There is also an optional rainful that can supress fire.

Drone Model Extension

To study the effectiveness that autonomous drones can contain wildfires, I made that into a new agent type. These drone agents represent aerial firefighting units capable of detecting and suppressing fires locally. At every tick, a drone scans in a fixed detection radius: if the fire is detected the drone moves towards it at a fixed speed to extinguish it within a specified supperssion radius. If no fire is detected, the drone patrols around the environment by turning with a random angle.

Some other behaviors include the total number of drones and their flying speed. To illustrate a decentralized behavior, the drones cannot communicate with each other nor retain memory. As a result, I can isolate the experiment on the local sensing rules themselves to discover any emerging patterns.

Finally, a stopping condition was added to the model, where the simulation ends when the fires are successfully extinguished— a condition that the original model misses, and by adding it would make the results much more compact and useful.

Experimental Design

To explore the emergent behaviors, I ran four different parameter sweeps across a variety of environmental and drone configurations in BehaviorSpace. These experiments were designed to isolate the following effects, each containing multiple stochastic runs per parameter:

1. Forest flammability and drone count
2. Wind presence and direction
3. Ember-driven “big jump” fire spread
4. Trade-offs between drone detection radius and suppression radius

Output Metrics

The output metrics capture both the overall damage and allows space to analyze the impact that drone coordination has through various stages of the burning process:

- Percentage of trees burned
- Burn rate over time
- Number of active fires
- Time to containment or system collapse

Results

General Patterns

Overall, the effectiveness of the fire containment was sensitive to both the environmental conditions and the drone parameter choices, with particular ranges inducing a greater percentage of success, with abrupt transitions between effective and ineffective parameter settings.

Discovery 1: Failure of Long-Range Detection

Figure 1 illustrates that larger detection radii do not improve performance. When increasing the number of drones, the detection radius of 20 consistently has worse outcomes than a detection radius of 10. This suggests that long-range detection could potentially lead to drones converging prematurely on the same fires, leading to clustering that reduces effective spatial coverage of the overall swarm. This gives space for fire to grow unchecked elsewhere, leading to a higher burn rate. On the other hand, an intermediate detection range balances this exploration-exploitation trade-off, hence leading to better outcomes across different experiments.

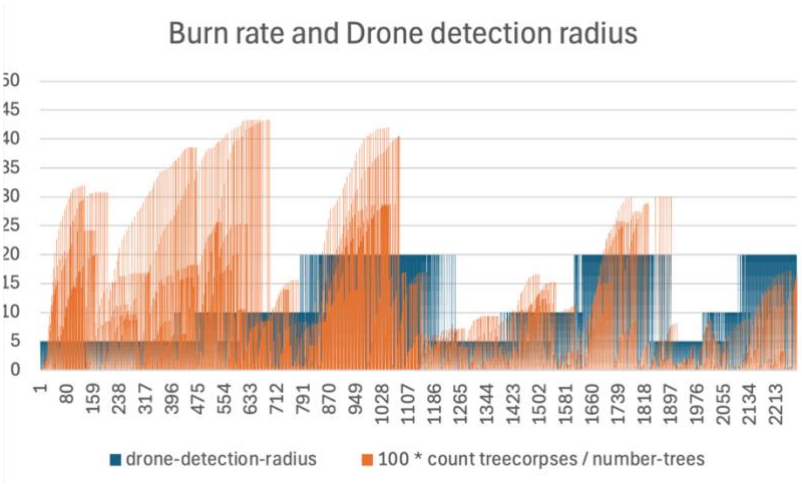


Figure 1

Discovery 2: Late-Stage Flare-Ups

A second emergent phenomenon was the appearance of late-stage flare-ups, where there is a temporary increase in burn rate right before the extinguishment of all the fire agents. This behavior is particularly obvious when there are less drones. One hypothesis is that these flare-ups are led by the exploitation-exploration trade-off too. When the drones concentrate heavily on suppressing existing clusters, they could potentially neglect further ignitions caused by either the "big jumps" or just random ignitions. As a result, even though the main fire cluster is almost extinguished, these neglected ignitions would have appear to have a slow "flare-up" at the end before the drones can detect, move, and extinguish them.

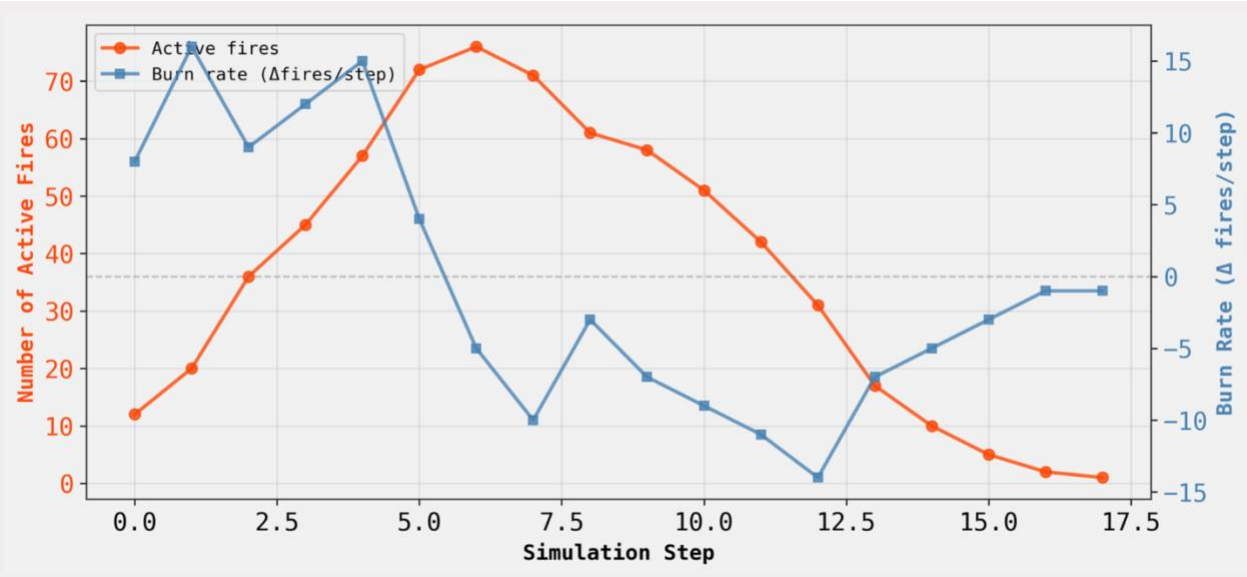


Figure 2

Discussion and Conclusions

This study demonstrates that a rule-based system for decentralized drone swarms are unpredictable depending on the parameter and environmentla contexts. However, one insight is that intermediate parameter values often outperform extreme ones, where overly aggressive parameters in sensing radius lead to convergence and loss of coverage, while overly local sensing leaves fires undetected.

These results highlight the vulnerability of purely local rules to exploration–exploitation trade-offs. In terms of practical implications for real-world drone swarm design, there are still a lot more to explore. The most immediate experiment could be including different ratios to test out whether the 'golden ratio' of exploration-exploitation applies to drone swarm coordination. Moreover, future models can implement more mixed strategies, introduce inter-drone communication, adaptive parameter responses, and model more realistic environments after specific terrains or regions prone to wildfires (such as California)

Bibliography

P. L. Andrews, C.D. Bevins, R.C. Seli. “BehavePlus fire modeling system, version 2.0: Users Guide ”. *USDA Forest Service Gen. Techn. Rep. RMRS-GTR-106WWW. Ogden*, 2003, 45 p.

F. Yang, Q. Yang, P. Wang, “Modelling for Forest Fire Evolution Based on the Energy Accumulation and Release ”. *International Journal of Advanced Computer Science and Applications*, Vol. 6, No. 9, 2015, pp. 38–45.

P. S. Shatalov, G. A. Dorrer “Parallel computation of forest fire and its interaction with infrastructure objects ”. *Fourth Fire Behave and Fuels Conference, conference abstracts. st. Petersburg.* 2013, p. 3.